Albert Poulose

Postdoctoral Researcher | Microgrid Research Center, Dept. of Electrical Engineering, Kyungpook National University, Daegu, South Korea
Email: albertpoulosepalatty@gmail.com | Website: albertpoulose.com | ORCID: 0000-0002-6138-7857 | LinkedIn: albert-poulose

Research profile

- Postdoctoral researcher in **power-system dynamic security** and **scientific machine learning**, focused on transient stability, converter-dominated grids (GFM/GFL), and physics-informed surrogates for trajectory prediction and critical clearing time (CCT) estimation.
- **Research areas:** Power-system stability | GFM/GFL converters | Scientific AI/PINNs | Digital twins
- **Key expertise:** Transient stability | Converter-dominated grids | Synchronizing torque & energy functions | PINNs | Parameter estimation | Python/PyTorch | MATLAB/Simulink | PSCAD

Selected research contributions

- **Physics-informed transient stability surrogate** (IEEE Access, 2026)
  Developed a PINN-based trajectory surrogate for TSA and CCT estimation on an SMIB benchmark; foundation for multimachine and parameter-estimation extensions.
- **Converter transient stability** (IEEE Access, 2024)
  Developed a transient-energy-function perspective for direct TSA of grid-connected voltage-source converters (GFM/GFL-relevant settings).
- **Synchronizing-torque stability metric** (Energies, 2022)
  Proposed a synchronizing-torque-based transient stability index for multimachine interconnected systems.
- **Renewable-rich grid stability** (Energies, 2023)
  Surveyed and structured transient stability analysis and enhancement techniques for renewable-rich power grids.

Research experience

2025.03 to Present

- **Postdoctoral Researcher** : **Kyungpook National University**, Daegu, South Korea
  Microgrid Research Center, Department of Electrical Engineering
  - Developing physics-informed dynamic surrogates for TSA and CCT estimation; extending toward multimachine systems and parameter estimation.
  - Converter-dominated and low-inertia stability themes (GFM/GFL) with Python-centric dynamic simulation workflows.
  - Collaboration with the Smart Power Systems Laboratory; mentoring junior researchers within the group.

2020.03 to 2025.02

- **Graduate Research Assistant** (Ph.D., full-time) : **Kyungpook National University**, Daegu, South Korea
  Smart Power Systems Lab, School of Electronic and Electrical Engineering
  Supervisor: Prof. Soobae Kim
  - Thesis on multifaceted TSA: synchronizing torque in multimachine systems and energy-function methods for renewable-connected converters.
  - Lyapunov/TEF analysis with MATLAB/Simulink and Python simulation for conventional and converter-dominated grids.

2017.08 to 2019.06

- **M.Tech research** (Power Systems) : Government Engineering College, Thrissur (affiliated with APJ Abdul Kalam Technological University), India
  Higher-order sliding-mode / super-twisting load frequency control for a two-area interconnected system.

Publications

4 first-author journal articles (3 during Ph.D.; 1 during postdoc) | 3 conference papers | Google Scholar: 59 citations, h-index 4, i10-index 3 (as of Sep. 2026).

Journal Articles

J1. A. Poulose and S. Kim, "Unified physics-informed trajectory surrogate for transient stability assessment and critical clearing time estimation on a single-machine infinite-bus benchmark," *IEEE Access*, vol. 14, pp. 122 177–122 196, 2026. DOI: 10.1109/ACCESS.2026.3718419

J2. A. Poulose and S. Kim, "Direct transient stability assessment of grid-connected voltage source converters: A transient energy functions perspective," *IEEE Access*, vol. 12, pp. 133 545–133 556, 2024. DOI: 10.1109/ACCESS.2024.3450316

J3. A. Poulose and S. Kim, "Transient stability analysis and enhancement techniques of renewable-rich power grids," *Energies*, vol. 16, no. 5, 2023, ISSN: 1996-1073. DOI: `10.3390/en16052495`

J4. A. Poulose and S. Kim, "Synchronizing torque-based transient stability index of a multimachine interconnected power system," *Energies*, vol. 15, no. 9, 2022, ISSN: 1996-1073. DOI: `10.3390/en15093432`

Conference Proceedings

C1. A. Poulose and S. Kim, "A technical briefing on machine learning for transient stability assessment in power systems," in *2025 International Conference on Advanced Information Technology (ADINTECH)*, Phnom Penh, Cambodia, 2025, pp. 54–60. URL: `https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12333641`

C2. A. Poulose and R. Kumar P., "Performance analysis of load frequency control for a two-area interconnected power system using twisting control," in *2019 International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2019. URL: `https://sites.google.com/view/albertpoulose/publications?authuser=0`

C3. A. Poulose and R. Kumar P., "Super-twisting algorithm based load frequency control of a two-area interconnected power system," in *2019 20th International Conference on Intelligent System Application to Power Systems (ISAP)*, New Delhi, India, 2019, pp. 1–5. DOI: `10.1109/ISAP48318.2019.9065992`

Research projects

2025.03 to Present
- **Physics-Informed Dynamic Security Assessment**
  Physics-informed trajectory surrogates for TSA and CCT estimation; multimachine extensions; inertia/damping estimation and system identification.

2022.03 to Present
- **Converter-Dominated Grid Stability**
  GFM/GFL interactions, weak-grid and low-inertia stability, and transient-energy methods for renewable-rich converter-connected systems.

2020.03 to 2025.02
- **Ph.D. research: Multifaceted transient stability analysis**
  Supervisor: Prof. Soobae Kim, Kyungpook National University
  Synchronizing-torque-based stability indices in multimachine systems; Lyapunov and transient energy-function methods for renewable-connected grids with GFM/GFL converters.

Current research directions

- **Physics-informed dynamic security:** Trajectory prediction, CCT estimation, and multimachine TSA with physics-informed surrogates.
- **Dynamic parameter estimation:** Inertia/damping estimation, system identification, and measurement-informed modeling.
- **Converter-dominated grids:** GFM/GFL dynamics, low-inertia systems, stability limits, and digital-twin applications.

Education

2020.03 to 2025.02
- **Ph.D., Electronic and Electrical Engineering** : **Kyungpook National University**, Daegu, South Korea
  School of Electronic and Electrical Engineering
  Thesis: Multifaceted Transient Stability Analysis of Power Systems: A Study of Synchronizing Torque in Conventional Systems and Energy Functions in Renewable Connected Systems
  Advisor: Prof. Soobae Kim
  GPA: 4.18/4.5

2017.08 to 2019.06
- **M.Tech., Power Systems** : **APJ Abdul Kalam Technological University**, Kerala, India
  Government Engineering College, Thrissur
  GPA: 7.84/10
  Thesis: Higher-order Sliding Mode Based Load Frequency Control for a Two-area Interconnected Power System

2011.07 to 2015.05
- **B.Tech., Electrical and Electronics Engineering** : **Mahatma Gandhi University**, Kerala, India
  SCMS School of Engineering and Technology, Ernakulam

Technical expertise

- **Research expertise:** Transient stability; small-signal stability; dynamic security; renewable integration; converter-dominated grids; low-inertia systems.
- **Converter dynamics:** GFM/GFL; VSM; PLL; droop control; weak-grid stability.

Technical expertise (continued)

- **Scientific AI:** PINNs; trajectory surrogates; hybrid physics-data modeling.
- **System identification & estimation:** Dynamic parameter estimation; inertia/damping estimation; measurement-informed modeling.
- **Simulation:** MATLAB/Simulink; PSCAD; PSS/E; PowerWorld; ANDES; PSAT; PST.
- **Programming:** Python; MATLAB; PyTorch; NumPy; SciPy; scikit-learn.

Languages
- English (proficient), Malayalam (native), Korean (intermediate), Hindi (beginner), Tamil (beginner)

Research software and computational work

- Python/PyTorch physics-informed workflows for trajectory prediction, TSA screening, and CCT estimation; dynamic simulation pipelines for SMIB and multimachine systems; CCT computation and stability screening; ANDES- and MATLAB/Simulink-based modeling for conventional and converter-dominated grids; reproducible workflows for simulation, dataset construction, and surrogate training.

Selected teaching

- University teaching concurrent with research: Logic Circuits (69 students) and Numerical Analysis (26 students) as Lecturer at KNU (2025–2026); Linear Control Systems and Electric Hybrid Vehicles as Assistant Professor at CCET (2019); UG TA support during M.Tech. Mentoring of junior researchers within the laboratory.

Awards and achievements

2020.03
- **KNU International Graduate Scholarships (KINGS)**: Tuition fee waiver for two years of Ph.D. coursework.

2016.03
- **Graduate Aptitude Test in Engineering (GATE)**: Qualified GATE-2016 (score 411; 90.90th percentile); M.Tech. stipend via AICTE (India).

References

Ph.D. Supervisor
- **Prof. Soobae Kim**
  Kyungpook National University
  School of Electronic and Electrical Engineering, Buk-gu, Daegu-41566, South Korea
  `soobae.kim@gmail.com`

M.Tech. Supervisor
- **Dr. Ramesh Kumar P.** , Associate Professor
  Government Engineering College, Kozhikode, Kerala, India
  APJ Abdul Kalam Technological University
  `rameshp36@gmail.com`